dimensions (to be discussed shortly) is true, and if these curled-up dimensions were themselves to expand to a macroscopic size, the lower-dimensional examples discussed make it clear that life as we know it would change immensely.

Surprisingly, though, even if they should always stay curled up and small, the existence of extra curled-up dimensions has profound implications.

Unification in Higher Dimensions

Although Kaluza's 1919 suggestion that our universe might have more spatial dimensions than those of which we are directly aware was a remarkable possibility in its own right, something else really made it compelling. Einstein had formulated general relativity in the familiar setting of a universe with three spatial dimensions and one time dimension. The mathematical formalism of his theory, however, could be extended fairly directly to write down analogous equations for a universe with additional space dimensions. Under the "modest" assumption of one extra space dimension, Kaluza carried out the mathematical analysis and explicitly derived the new equations.

He found that in the revised formulation the equations pertaining to the three ordinary dimensions were essentially identical to Einstein's. But because he included an extra space dimension, not surprisingly Kaluza found extra equations beyond those Einstein originally derived. After studying the extra equations associated with the new dimension, Kaluza realized that something amazing was going on. The extra equations were none other than those Maxwell had written down in the 1880s for describing the electromagnetic force! By adding another space dimension, Kaluza had united Einstein's theory of gravity with Maxwell's theory of light.

Before Kaluza's suggestion, gravity and electromagnetism were thought of as two unrelated forces; nothing had even hinted that there might be a relation between them. By having the bold creativity to imagine that our universe has an additional space dimension, Kaluza suggested that there was a deep connection, indeed. His theory argued that both gravity and electromagnetism are associated with ripples in the fabric of space. Gravity is carried by ripples in the familiar three space dimensions, while electromagnetism is carried by ripples involving the new, curled-up dimension.